

Dear Reviewers,

We greatly thank you for the constructive and insightful comments, which were very helpful in improving the clarity of our paper. In the revised version of the paper, we carefully addressed your comments. In the remainder of this letter, we provide a detailed list of changes highlighted in blue as a response to each comment.

Best regards,

Engelbert Fellingner and Patricia Lago

Please find the reviewer feedback below:

SUBMISSION: 20

TITLE: Assessing the Maturity of Sustainability-aware Data Management Strategies that minimise Data Volumes and reduce Energy Consumption

----- REVIEW 1 -----

SUBMISSION: 20

TITLE: Assessing the Maturity of Sustainability-aware Data Management Strategies that minimise Data Volumes and reduce Energy Consumption

----- Overall evaluation -----

SCORE: 1 (weak accept)

----- TEXT:

This paper addresses a timely and relevant topic by proposing a study design for developing a sustainability-aware data management maturity assessment model. The motivation is well articulated, and the planned mixed-method research design is comprehensive and appropriate for the stated research goals. I believe the paper is a valuable work-in-progress contribution. The following suggestions are intended to improve the clarity of the paper and help readers better understand the proposed research.

1. The first point concerns the title. The current title suggests that the paper presents an assessment of sustainability-aware data management maturity, whereas the main contribution is the design of a study to develop the SusDMMA-DVE model. A title that more explicitly reflects the methodological nature of the contribution would better align readers' expectations with the paper's actual scope.

We appreciate your recommendation. We have added the subtitle "...: A Study Approach" to the title to better reflect the methodological nature of the contribution and to align reader expectations with the actual scope of the paper.

2. Since the primary contribution is the proposed study design, the related work could also better position the methodological choices. At present, it mainly reviews existing DMMA literature but provides limited discussion of how such models are typically developed and validated. Including related work on comparable study designs (e.g., mixed-method

approaches or Design Science Research applications for maturity model development) would help readers understand which methodological elements are established practice and which constitute the paper's novel contribution (beyond the clearly added sustainability perspective).

Thank you for your feedback. In Section 2 (Related work), we have included the work of Mettler, which outlines the typical phases involved in the development and application of a maturity model from a design science research perspective. Using artificial and naturalistic evaluations, we incorporate DS into a mixed-method study design to define, iteratively refine and validate the goals, CSFs, KPIs, and maturity levels of our proposed model during development.

3. The paper would further benefit from a clearer presentation of the intended structure of the proposed maturity model. While the manuscript introduces goals, critical success factors (CSFs), KPIs, metrics, measurements, and maturity levels, it remains difficult to understand how these elements fit together in the final model. A conceptual overview, preliminary model, or illustrative example would greatly improve readability and make the proposed approach easier to follow.

Thank you for your comment. We have provided more clarity on the intended structure of the proposed maturity model through a description of its development process in Section 3. The expected outcomes are outlined in Section 3.6.

4. Related to this, the development process of the model components could be described more explicitly. The paper indicates that metrics will be defined during the case studies, but it is less clear when goals, CSFs, KPIs, and maturity levels are initially defined, iteratively refined, and finally validated. Clarifying this progression within the study design would strengthen the methodological transparency. Maybe they can also be placed on the fig. 1?

We have made the development process more distinct by detailing the steps and the desired results of Design Iteration 1 & 2 in Section 3. In addition, we have updated fig.1 accordingly.

5. The paper also builds upon the authors' previous work on sustainability-aware data management strategies and supporting or hindering factors. It would therefore be helpful to explicitly show how these previously established concepts are incorporated into the proposed model. Likewise, the role of the patterns (themes) identified through thematic analysis and their relationship to the model structure could be explained more clearly. Even a simplified conceptual figure illustrating these relationships would help readers understand how the different components contribute to the final assessment model.

Thank you for your comment. Our previous work informs the interviews and case studies that underpin the development of the proposed SusDMMA-DVE model. We have made these relationships explicit in the Introduction. The patterns identified will inform the iterative development of the model's questionnaire and maturity levels. Given the page limit, there was no space to include a figure.

6. Finally, throughout the manuscript, terms such as goals, aims, objectives, and purposes are used at different levels (research goals, model goals, study-step goals, and KPI-related

goals). A small editorial revision to make these distinctions more consistent would improve readability and reduce potential ambiguity.

Thank you; we have addressed this issue by removing synonyms that might confuse the reader.

These suggestions are intended primarily to improve the presentation and clarity of the proposed methodology rather than to change the underlying research. Addressing them would make the contribution easier for readers to understand and would better highlight the novelty and expected outcomes of the proposed study.

----- Language Quality and Readability -----

SCORE: 3 (fair)

----- REVIEW 2 -----

SUBMISSION: 20

TITLE: Assessing the Maturity of Sustainability-aware Data Management Strategies that minimise Data Volumes and reduce Energy Consumption

----- Overall evaluation -----

SCORE: 2 (accept)

----- TEXT:

Relevance: The topic of sustainability-aware data management is an important research direction and aligns well with the focus of the conference. The development of a maturity model that incorporates environmental considerations seems to be novel. But the authors could clarify the novelty of the proposed model in contrast to existing green-ICT models.

Thank you for your comment. Our approach differs from existing green-ICT models in that it focuses specifically on addressing two critical aspects: reducing data volumes and conserving energy resources while considering the four core sustainability dimensions. We have clarified this in our Introduction and in Section 2 (Related work).

Methodology: The manuscript presents a well-structured mixed-methods research plan that combines a systematic literature review, semi-structured interviews, case studies, and focus groups. The design follows the well-established guidelines and the intended iterative refinement of the model via DSR is a scientifically solid approach. The research question is clearly formulated.

Limitations: The authors have not provided any initial validation results or measurements, which would improve the paper. But for a short paper or work-in-progress paper this is completely acceptable, if the program committee accepts purely conceptual papers. The conclusion Section is a little bit brief and could be extended.

Thank you; we have extended Section 3.6 and the conclusion Section with more details on the model.

While a definition of sustainability-aware data-management maturity is provided, it should be differentiated from existing concepts more clearly. A more precise positioning of the contribution relative to prior work would be helpful.

Thank you for the comment; we have made our contribution more explicit in the Introduction.

Overall Impression: It outlines a promising research direction and addresses an important research gap. However, it currently does not deliver verifiable results, but this is acceptable as this short paper seems like an initial step towards a full journal paper.

Therefore, I recommend acceptance.

----- Language Quality and Readability -----

SCORE: 4 (good)

----- TEXT:

Overall, the paper is clear and well written.

I recommend increasing the font size in Tab. 1, which would make it easier to read.

Thank you for your recommendation; we have increased the font size to improve the readability of Tab. 1.

There is additional space between the Fig 1. and its caption, which could be removed.

We have removed the additional space.